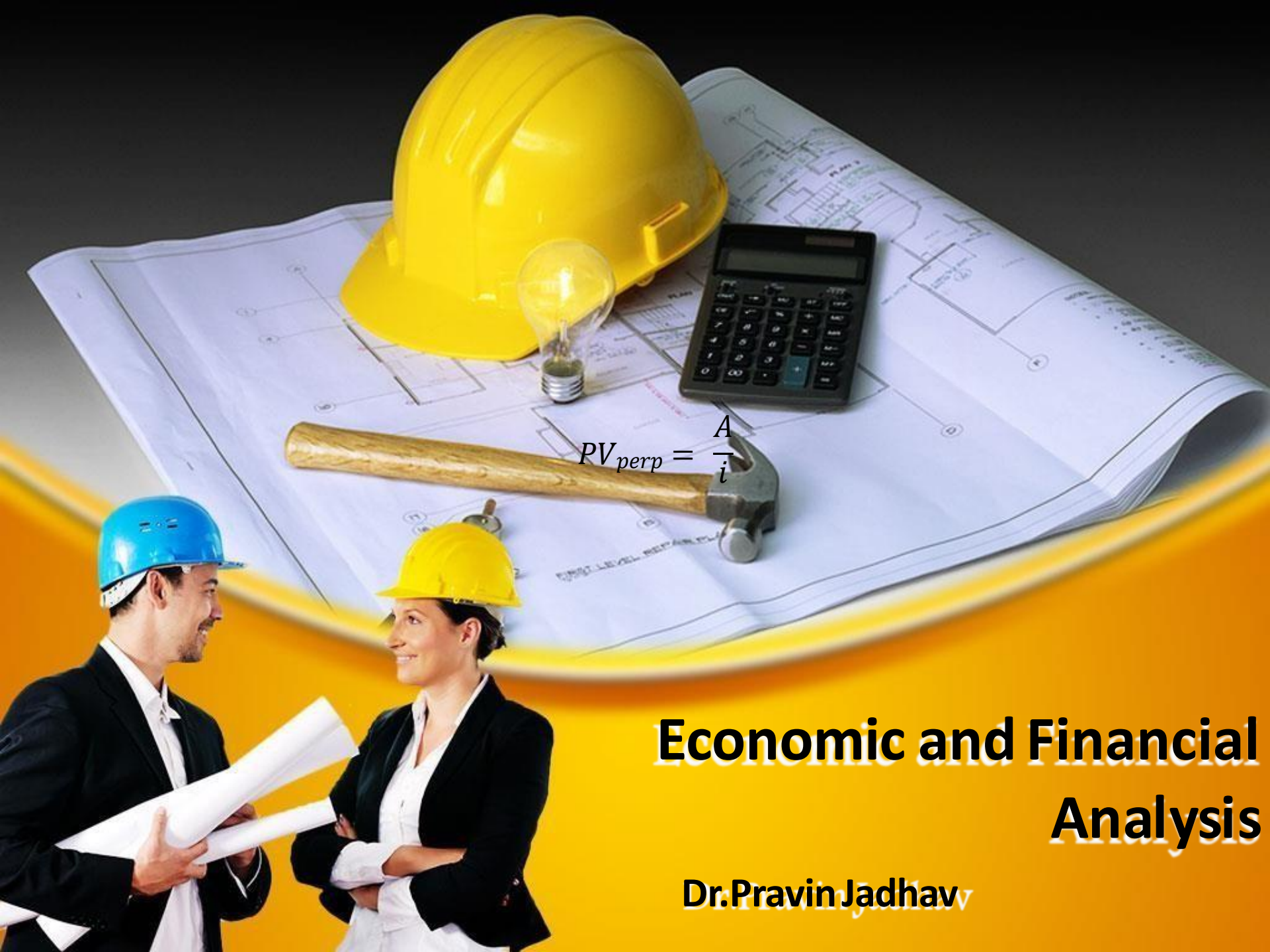




$$PV_{\text{perp}} = \frac{A}{i}$$

Economic and Financial Analysis

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Agenda



- *Project Viability*
- *Time Value of Money*
- *Cost of Capital*
- *Capital Budgeting Techniques*

Types of financial Analysis



- Estimates of investment (capital) cost and annual cost of the project in terms of monetary requirements
- Schedule showing the breakdown of the investment cost by years, with separate accounts of expenditure for construction and for the other categories of costs needed to bring the project into operation
- For the case of a developing country, estimates of portions of investment cost in domestic and foreign funds, especially when foreign currencies are in short supply.
- Plan for financing the costs of the project investment, including the sources of funds and the terms for repayment of each category of borrowing
- Estimates of costs, revenues from the sale of products and services; and required subsidies, on a year-by-year basis extending from the completion of construction to the date when repayment of all borrowed funds is completed, any beyond if appropriate.
- Plan for the required annual subsidies, if any, and for working funds to enable operation to commence and to meet temporary cash flow requirements during the early year of operation
- Additional statements of a financial nature depending on the regulatory agencies and financial institutions involved in the project.

Project Viability



- Technical Viability
- Economic or financial viability
- If a project is considered to be viable, then
$$\Sigma \text{ Benefits} > \Sigma \text{ Costs}$$
- How do measure benefits and costs
- Benefits are the profits earned by the project
- Costs are the investment needed to have assets in place

Time Value of Money



- TVM is based on the idea that money available at the present time is worth more than the same amount in the future due to its potential earning capacity.

Time Value of Money



- **Example:**
- If a give a choice between receiving Rs.100 today or Rs.100 a year after now, you should take the money today.
- Because, you could invest that Rs.100, and even if you only earned a 8% annual return on your investment, you still would have Rs.108 a year from now-obviously more than Rs.100 you would have received if you waited.
- if you didn't invest that Rs.100 at all but simply spent it, you would still be better off; because of inflation, the Rs.100 usually will have more buying power today than in the future. Clearly the first option is more valuable based on two fundamental concepts 1. Higher Purchasing Power and 2) better Opportunity cost.

Time Value of Money



- If 100 rupees today can be invested in bank at interest of 10% then after 1 year the value of 100 rupees =

$$100 * (1.1) = 110$$

At the end of two years, the value of 100 rupees would be

$$= 100 * (1.1) * (1.1)$$

$$= 100 * (1.1)^2 = 121$$

Using the same approach the present value of 100 rupees that would be received one year from now

$$= \frac{100}{1.1}$$

$$= 90.91$$

Formula..



- $FV_1 = PV * (1 + i)^1$
- $FV_2 = PV * (1 + i)^2$
- $FV_n = PV * (1 + i)^n$

- $PV = \frac{100}{1.1}$
 $= 90.91$

Application of Time Value



Period	1	2	3
Cash flow	100	200	300

$$\frac{100}{1.1} = 90.9$$

$$\frac{200}{1.1^2} = 165.2$$

$$\frac{300}{1.1^3} = 225.3$$

$$Total = 90.9 + 165.2 + 225.3 = 481.4$$

Example



- Consider a project that requires a Rs.4,00,000 investment today (a negative cash flow) and will return Rs.1,00,000 a year for the next five years (positive cash flows).
- $\text{Rs.4,00,000} = 1\text{st Year } 1,00,000 + 2\text{nd Year } 1,00,000 + 3\text{rd Year } 1,00,000 + 4\text{th Year } 1,00,000 + 5\text{th Year } 1,00,000 = \text{Rs.5,00,000}$
- On paper, it looks as if the project produces a Rs.1,00,000 profit. But those future cash flows must be converted to present value to know the actual profit. if the company uses a discount rate* of 10%,
- What is the present value of those cash flows?



$$PV = \frac{A}{(1+i)} + \frac{A}{(1+i)^1} + \dots + \frac{A}{(1+i)^n}$$

- actually comes out to Rs.3,79,078.52. That's less than the Rs.4,00,000 cost, so the project actually will lose money.
- However, if the company is using a discount rate of 5 percent, the present value is Rs.4,32,947.66 meaning the project is profitable.
- *The discount rate in DCF analysis takes into account not just the time value of money, but also the risk or uncertainty of future cash flows; the greater the uncertainty of future cash flows, the higher the discount rate

Same cash flow



Period	0	1	2	3	4	5
	A	A	A	A	A	A

$$PV = \frac{A}{(1+i)} + \frac{A}{(1+i)^1} + \dots \frac{A}{(1+i)^n}$$

$$a = \frac{A}{(1+i)} \quad r = \frac{1}{(1+i)}$$

$$A = \left[\frac{1 - \frac{1}{(1+i)^n}}{i} \right]$$

Capital Budget Techniques :

Pay Back Period



- Time needed to recover the initial cost outlay of the project
- Accept project if project payback period is less than the desired payback period
- Project A has following cash flow:

Year	0	1	2	3	4	5
Cash flow	-10,000	6000	4000	3000	2000	1000

Features of payback method



- Advantages
 - Easy to visualize, quickly understood, and easy to calculate
 - Eliminate Projects that do not provide returns in early years

Features of payback method



- Shortcomings
 - Does not use time value
 - There could be a certain arbitrariness in selecting the payback period
 - Does not consider cash flows that occur after the payback period

Capital Budget Techniques :

Pay Back Period



- Project A has following cash flow:

Year	0	1	2	3	4	5
Cash flow	-10,000	6000	4000	3000	2000	1000

- Project B has following cash flow :

Year	0	1	2	3	4	5
Cash flow	-10,000	5000	5000	0	0	0

Net Present Value



$$NPV = \sum_{i=1}^n \frac{Cash\ Flow_i}{(1 + r)^i} - Initial\ Investment$$

r – discount or interest rate

n – the number of time periods

i – the cash flow period

- NPV = (Sum of the present value of project cash flows) – (investment outlay)
- $NPV > 0$: Accept
- $NPV < 0$: Reject

Net Present Value : Illustration



- Project A has following cash flow, Calculate NPV considering Discount Rate as 8%

Year	0	1	2	3	4	5
Cash flow	-400000	70000	120000	140000	140000	40000

- 1) Calculate PV for all years
- 2) Sum the PV of all years
- 3) $NPV = (\text{Sum of the present value of project cash flows}) - (\text{investment outlay})$
- 4) If NPV is positive Accept the Project, if not Reject the project

NPV



Year	Cash flow	PV @ 8%
0	(-400000)	
1	70000	64814.81
2	120000	102880.7
3	140000	111136.5
4	140000	102904.2
5	40000	27223.33
6	Sum of PV	408959.5
7	NPV	8959.494

Example



Year	Cash flow	PV @ 8% DR	PV @ 15% DR
0	(-400000)		
1	70000	64814.81481	60869.56522
2	120000	102880.6584	90737.24008
3	140000	111136.5137	92052.27254
4	140000	102904.1794	80045.45438
5	40000	27223.32788	19887.06941
6	Sum of PV	408959.4943	343591.6016
7	NPV	8959.494267	-56408.39837



Adani Infrastructure Company is planning to undertake an infrastructure project with initial investment of ₹2,40,000.

Consider the following cash flow from the project and suggest whether Adani Infrastructure should invest at 10%, 15% and 20% discount rate.

Why and Why not?

Year	2019	2020	2021	2022
Cash flow	₹ 25,000	₹ 75,000	₹1,50,000	₹ 1,50,000

Example



Reliance Infrastructure has four potential projects all with an initial cost of ₹1,500,000. The capital budget for the year will only allow Reliance Infrastructure to accept one of the four projects. Given the discount rates and the future cash flows of each project, which project should they accept according to Net Present Value (NPV) of project?

Cash Flows	Project A	Project B	Project C	Project D
Year one	₹ 3,50,000	₹ 4,00,000	₹ 7,00,000	₹ 2,00,000
Year two	₹ 3,50,000	₹ 4,00,000	₹ 6,00,000	₹ 4,00,000
Year three	₹ 3,50,000	₹ 4,00,000	₹ 5,00,000	₹ 6,00,000
Year four	₹ 3,50,000	₹ 4,00,000	₹ 4,00,000	₹ 8,00,000
Year five	₹ 3,50,000	₹ 4,00,000	₹ 3,00,000	₹ 10,00,000
Discount Rate	4%	8%	13%	18%

Profitability Index



- $PI = (\text{Sum of the PV of project cash flow}) / (\text{Initial Outlay})$
- $PI > 1$: Accept
- $PI < 1$: Reject

Internal Rate of Return



- IRR is a discount rate at which NPV becomes zero
- In other words, IRR is the opportunity cost at which the NPV becomes zero.
- IRR as the name suggests, it tells how much rate of return are we getting from the project
- Use of calculating IRR
- 1) IRR is used to rank different project
- The higher a projects IRR the more desirable it is to undertake the project
- If all the other factors are same for differrent projects then the projet with highest IRR should be considered

Internal Rate of Return



- IRR gives the rate of return that the project earns
- It is the discounted rate that equates the present value of the projects future cash flow with projects initial outlay
- $IRR > \text{Required rate of return} : \text{Accept}$
- $IRR < \text{Required rate of return} : \text{Reject}$

IRR Formula



$$0 = \text{NPV} = \sum_{t=1}^T \frac{C_t}{(1 + \text{IRR})^t} - C_0$$

where:

C_t =Net cash inflow during the period t

C_0 =Total initial investment costs

IRR =The internal rate of return

t =The number of time periods

Illustration - IRR



- Project has the following cash flow

Year	0	1	2	3	4
Cash flow	-100	20	30	40	45

- NPV @ 15% = 0.800
- NPV @ 16% = -1.36
- IRR = 15.37%



$$IO = \sum_{t=1}^n \frac{CF_t}{(1 + IRR)^t}$$

$$100 = \frac{20}{(1 + IRR)} + \frac{30}{(1 + IRR)^2}$$

Application of Time Value

– Loan Amortisation



- Amortised loans : When loans are paid in equal periodic installments
- E.g
- Loan – 1,00,000
- 5 equal instalments
- $i = 15\%$
- $EMI = P \times r \times (1 + r)^n / ((1 + r)^n - 1)$ where
P= Loan amount, r= interest rate,
n=tenure in number of months.



$$A / \left[\frac{1 - \frac{1}{(1.15)^5}}{0.15} \right]$$

$$A = 29831$$

Loan Amortisation- Table



Year	Loan	Annual Payment	Interest	Principle Repay	Remaining Balance
1	100000	29831	15000	14831	85169
2	85169	29831	12775.35	17055.65	68113.35
3	68113	29831	10216.95	19614.05	48498.95
4	48499	29831	7274.85	22556.15	25942.85
5	25942	29831	3891.3	25939.7	2.3 ~ 0

Importance of Amortization Table



- Interest rate –Interest Coverage Ratio

$$ICR = \frac{EBIT}{Interest\ Expense}$$

- How easily a company can pay their **interest** expenses on outstanding debt.
- Important leverage indicator-what is margin that the company has in terms of paying the interest
- Interest qualifies for tax exemption
(EBIT – Earning before Interest and Tax)

Perpetuity Cash flow



- Cash flow that occur for a indefinite period of time.
- Example – Pension, Airport

Perpetuity Cash flow



$$PV = A * \left[\frac{1 - \frac{1}{(1+i)^n}}{i} \right] \qquad PV_{per} = \frac{A}{i}$$

$$PV_{perp\ growing} = \frac{A * (1 + g)}{(i - g)}$$

Cost of Capital



$$PV = \frac{FV_t}{(1 + k)^t}$$

k = Discount Rate (Cost of Capital)

t = time of cash flow

What determines cost of capital –project features



- Project Risk
 - Uncertainty in cash flow
- Duration of investment
 - Long term vs Short term investment
 - Risk and Return are directly prop..

What determines cost of capital- Source of Capital



- Source of investment
 - Broadly two types of sources : Equity and debt
- Equity – is more expensive
- Debt- is cheaper

Cost of debt



- Determining cost of debt is easier
- Interest rate on debt indicates the return expected by the debt investor from investing in the project
- Cost of debt is nothing but the interest rate on debt
- Interest payments qualify for tax shields
- We should therefore use cost of debt on an after-tax basis in our calculations

Example



- Project – Rs. 1000/-

	Company A	Company B
Equity	1000	200
Debt		800
Total	1000	1000
Revenue	1000	1000
Expenditure	500	500
PBIT (profit before interest and tax)	500	500
PBT (profit before tax)	500	420
PAT (profit after tax)	250	210

Example



- Interest – 10%
- Tax -50%
- Company A – 250
- Company B – 210
- Tax saving – 40
- Net interest = 80-40
- Interest after tax = $40/800 = 5\%$



- $5\% = 10\% * (1 - \text{tax rate})$
- $k_d(\text{post tax}) = k_d * (1 - \text{tax rate})$
- *$k_d = \text{cost of debt}$*

Cost of equity



- Equity does not involve contractual payment obligations
- Estimating the cost of equity differs from the procedure for determining the cost of debt
- A common and useful model for estimating the cost of equity can be done by the Capital Asset Pricing model



$$k_e = r_f + \beta * (r_m - r_f)$$

- Where,
- K_e = Cost of equity
- r_f = risk free rate of investment
- β = Risk of equity as compared to the market
- r_m = Return on Market portfolio

Weighted Average Cost of Capital



- It is usual to have an investment to be funded by both debt and equity
- The cost of capital i.e. the rate used to discount the future cash flows should therefore be a composite of both equity and debt costs
- It is referred as the Weighted Average Cost of Capital (WACC)



$$k_{wacc} = k_d * (1 - t) * \frac{D}{D+E} + k_e * \frac{E}{D+E}$$

- weighted average cost of capital is nothing but cost of debt on an after-tax basis multiplied by the proportion of debt funding that we use for funding the project.
- So, let us assume D is the amount of debt that we have used.
- The total funding for the project it is nothing but the debt plus equity.
- So, this ratio D by D plus E, represents the proportion of debt funding that we actually have to use for the project plus cost of equity multiplied by the proportion of equity funding.
- In essence this is nothing but a weighted average cost of both debt and equity, that is simply the weighted average cost of capital.

Illustration : WACC calculation



- Risk free interest rate : 8%
- Equity beta : 1.2
- Market Premium : 8.4%
- Pre- tax cost of debt : 10%
- Proportion of debt financing : 80%
- Income tax rate : 35%



- After tax cost of debt = $k_d * (1 - \text{tax rate})$
= $10\% * (1 - 35\%)$
= 6.5%



Cost of equity $k_e = r_f + \beta * (r_m - r_f)$

$$= 8\% + 1.2 * 8.4\% = 18.08\%$$



$$k_{w a c c} = k_d * (1 - t) * \frac{D}{D+E} + k * \frac{E}{D+E}$$

$$= 6.5\% * 80\% + 18.08\% * 20\%$$

$$= 8.82\%$$

Thought Question??



- Why is debt generally cheaper than equity?
- How do we get the values of equity and debt for calculating the WACC?

Thought Questions



- What are the shortcomings of using IRR?
- What are the assumptions behind the IRR calculation?

Guidelines for Capital Budgeting



- Identification of project cash flow
- Project cash flows fall into the following categories
 - Initial Outlay
 - Annual free cash flows
 - Changes to working capital
 - Terminal cash flow

Guideline for Capital Budgeting



- Only incremental cash flows matter
- ✓ Will this cash flow occur if the project is accepted?
- ✓ Will this cash flow occur if the project is rejected?
- Account for opportunity costs
- Working capital requirements have to be considered
- Ignore sunk cost
- Ignore interest payments and financing cash flow

Initial Outlay



- Investment needed till the project begins operation
 - Equipment
 - Installation
 - Construction

Operating Cash flow



- Cash flow from operations (OCF)
 - $OCF = \text{Net Income} + \text{Depreciation}$



- Revenues – 1000
- Expenses – 500
- Depreciation – 150
- EBIT – 350
- Tax 34% - 110
- PAT - 231

Changes in working capital



- Increase in net working capital as treated as a cash flow
- When company is increasing the inventory levels, the current assets increases, as it need to pay more for accumulating that inventory
- Increase in current liabilities or reduction network capital is treated as cash inflow
- It is assumed upon termination of project, working capital is liquidated to cash at book value

Terminal Value



- Is the cash flow from disposing the assets upon termination of the project
- We can liquidate all the assets and treated it as a cash inflow

Example – Calculating Cash flows



- Project A has the following projections for unit sales

Year	1	2	3	4	5
Unit Sales	50000	100000	100000	70000	50000
Price Per Unit	150	150	150	150	130

- Cost of new plant, equipment and installation – 10 million
- Variable Cost – 80/Unit
- Fixed Costs – 5,00,000 per year
- Working Capital – 100000 to get production started. For each year investment will be 10% of sales in the year
- Straight line depreciation, with no salvage value
- Tax Rate – 34%
- Cost of Capital = 15%



- Year 1
- Revenue – $50000 \times 150 = 7500K$
- Cost = $50000 \times 80 + 500K$
= 4500 K
Dep = 2000K
EBIT = 1000K
Tax = 340 K
PAT = 660 K

Operating Cash flow



- Year 1 - 2660
- Year 2 – 4970
- Year 3 – 4970
- Year 4 – 3584
- Year 5 - 2000

Changes to working capital



- Initially we have to incur an investment in working capital of 100000 and every year working capital investment is 10% of your sales
- Year 1 sales is 7.5 million therefore working capital investment in year 1 will be 750 K
- Increase in working capital : reduction in cash flow
- Working capital increases from 100000 to 750000 to year 1 so difference is 650000 so it is cash outflow

Terminal value



- Terminal value, we assume to be 0, because the assets do not have any salvage value.
- Upon disposal on project termination, we get nothing, so in this case, the project terminal value at the end of project life become 0



Year	0	1	2	3	4	5
Initial Outlay	-10,000					
OCF		2660	4970	4970	3584	2000
Change in WC	-100	-650	-750	0	450	1050
Cash Flow	-10,100	2,010	4,220	4,970	4,034	3,050
NPV	1933					



Year	0	1	2	3	4	5
Cash Flow	-10100	2010	4220	4970	4034	3050
IRR	22%					

Example – Infrastructure Project



- Cogeneration power plant, 250 MW capacity
- Plant functions at 90% capacity
- Prices
 - Electricity – 40/mega-watt hour, 6% annual increase
 - Steam – 4/- thousand pounds, 5% annual increase
 - Natural Gas – 3/million BTU , 6% annual increase

	At capacity	Maximum Annual	90% utilization
Electricity Production	250 MW	2,190,000 MWH	1,971,000 MWH
Steam Production	150,000 PPH	1314MP	1,182.6MP
Natural Gas	1.950 MBTU/ hour	17,082 B BTU	15,373.8 B BTU

Example – Infrastructure Project



- Operating and other cash expenses
 - First year = 8 million per year; 5% annual increase
- Tax Rate – 40%
- Assume negligible investment are needed for working capital requirements
- Total investment needed : 113,508 million; 25% equity; 75% debt
- Straight line depreciation for 10 years

Example – Infrastructure Project



- Loan amount – 85,131 million
- Term – 10 years
- Interest Rate – 10% per annum
- Principle repayment
 - Years 1-3 : 5%
 - Years 4-7 : 10%
 - Years 8-10 : 15%
- Cost of equity – 29%

Steps Involved



- Calculate the different cash flows
 - Initial Outlay : 113,508 million
 - Operation cash flow

Operating Cash Flow



$$P_1 = \frac{10000000 \text{ MWh} \times 10/\text{MWh}}{1000000} = 78.84$$

- $P_2 = 78.84 \times 1.06 = 83.57$
- $P_3 = 83.57 \times 1.06 = 88.58$

Operating Cash Flow



$$S_1 = \frac{1102000 \cdot 11^{-1} \cdot 1/11}{1000000} = 4.73$$

- $S_2 = 4.73 \cdot 1.05 = 4.97$
- $S_3 = 4.97 \cdot 1.05 = 5.22$

Operating Cash Flows



Operating cash flows

Year	Revenues		Cash expenses		Non cash expenses	Pre-tax income	Post tax income	Cash flow from operations
	Electricity	Steam	Natural gas	Operating				
1	78.84	4.73	46.12	8.00	11.35	18.10	10.86	22.21
2	83.57	4.97	48.89	8.40	11.35	19.90	11.94	23.29
3	88.58	5.22	51.82	8.82	11.35	21.81	13.08	24.43
4	93.90	5.48	54.93	9.26	11.35	23.83	14.30	25.65
5	99.53	5.75	58.23	9.72	11.35	25.98	15.59	26.94
6	105.51	6.04	61.72	10.21	11.35	28.25	16.96	28.31
7	111.84	6.34	65.42	10.72	11.35	30.68	18.41	29.76
8	118.55	6.66	69.35	11.26	11.35	33.25	19.95	31.30
9	125.66	6.99	73.51	11.82	11.35	35.97	21.58	32.93
10	133.20	7.34	77.92	12.41	11.35	38.85	23.31	34.66
11	141.19	7.71	82.60	13.03		53.27	31.96	31.96
12	149.66	8.09	87.55	13.68		56.52	33.91	33.91
13	158.64	8.50	92.81	14.37		59.95	35.98	35.98
14	168.16	8.92	98.37	15.09		63.62	38.17	38.17
15	178.25	9.37	104.28	15.84		67.50	40.50	40.50



Changes in Working Capital



- This project we assume that working capital investments are negligible, so therefore, there are no there are no values that needs to be considered for changes to working capital investment.

Terminal or Residual value



$CF_{15}, CF_{16}, CF_{17}, \dots$

$$P_{15} = \frac{CF_{16} * (1+g)}{(k-g)} \quad \text{where, } k = \text{cost of capital and } g = \text{growth rate}$$

Assume cost of capital is 3 %

Calculate Cost of Capital by WACC = cost of debt (75%) + cost of equity (25%) = 12%

$$= \frac{45 * 1.03}{12\% - 3\%} = 490.76$$

NPV & IRR



- Net Present Value of 150.32.
- IRR = 26%
- Cost of Capital = 12%

Mutually Exclusive Projects



- Conflicting recommendations could be because of
 - Size differences
 - Cash flow timings

Size differences



- Project A : IRR-30%, NPV -100
- Project B : IRR- 24%, NPV -200
- Under such circumstances, select the project that add greatest wealth

Cash flow timing differences



- Lets consider two projects with differences in cash flow timing

Year	0	1	2	3	4	5	6	IRR	NPV
Project A	-200	80	80	60	60	40	20	22%	₹ 55.48
Project B	-200	40	40	60	80	80	100	20%	₹ 68.42